Multiple Linear Regression

Multiple inputs and comparable feature scales

Hand-calculation worksheet

Name _____ Date _____

01 Data and Features

* Synthetic practice data Area is entered in units of 10 m^2 · For 120 m^2, use $x_1 = 12$

House i	Area (m^2)	Input x_1	Rooms x_2	Target y (10M KRW)
1	120	12	3	21
2	60	6	2	12
3	160	16	4	28
4	100	10	3	22
5	180	18	4	31
6	80	8	2	15
7	140	14	3	25

1 Write the number of examples m and features n

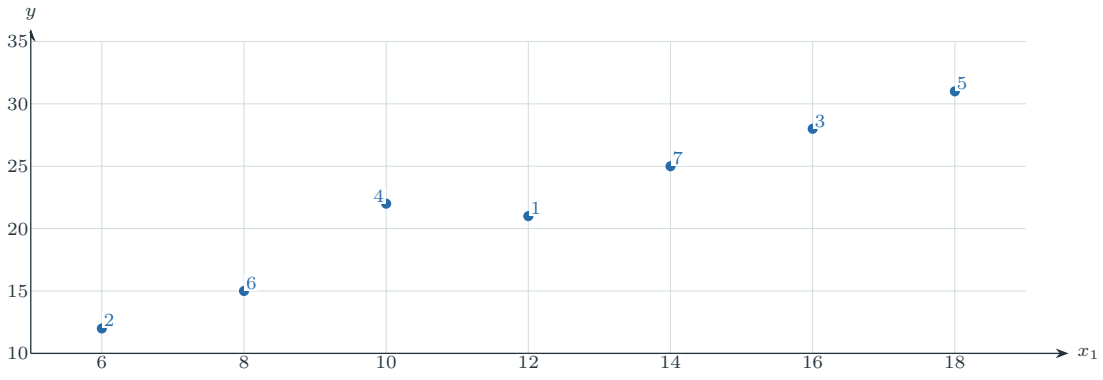
$m =$ _____ $n =$ _____

2 Express the input of House 4 as a vector and find the following values

$\mathbf{x}^{(i)} = [x_1^{(i)}, x_2^{(i)}]$ i = house index, j = feature index

$\mathbf{x}^{(4)} = [$ _____ , _____ $]$ $x_2^{(4)} =$ _____ $y^{(4)} =$ _____

3 Is a larger house always more expensive in the actual data? Compare Houses 1 and 4



☐ Always more expensive ☐ Not always

Reference · DeepLearning.AI / Stanford Online · Machine Learning Specialization · Current lesson slides 83 to 99

02 Prediction and Cost

$f_{\mathbf{w},b}(\mathbf{x}) = w_1x_1 + w_2x_2 + b \qquad \mathbf{w} = [1, 3], \quad b = 2$

4 Calculate the price of House 4 using both the expanded expression and a dot product

$\mathbf{w} \cdot \mathbf{x} = w_1x_1 + w_2x_2 \qquad f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$

$1 \times 10 + 3 \times 3 + 2 =$ _____

$[1, 3] \cdot [10, 3] + 2 =$ _____

5 Calculate each of the seven predictions, errors, and squared errors

$\hat{y}^{(i)} = x_1^{(i)} + 3x_2^{(i)} + 2 \qquad e^{(i)} = \hat{y}^{(i)} - y^{(i)}$

House	Prediction expression	$y^{(i)}$	$\hat{y}^{(i)}$	$e^{(i)}$	$(e^{(i)})^2$
1	$12 + 3 \times 3 + 2$	21			
2	$6 + 3 \times 2 + 2$	12			
3	$16 + 3 \times 4 + 2$	28			
4	$10 + 3 \times 3 + 2$	22			
5	$18 + 3 \times 4 + 2$	31			
6	$8 + 3 \times 2 + 2$	15			
7	$14 + 3 \times 3 + 2$	25			

6 Calculate the MSE and cost

$\text{MSE} = \frac{1}{m}\sum_{i=1}^{m}(e^{(i)})^2 \qquad J(\mathbf{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}(e^{(i)})^2 = \frac{1}{2}\text{MSE}$

$\sum_{i=1}^{7}(e^{(i)})^2 =$ _____

MSE = _____ $J(\mathbf{w}, b) =$ _____

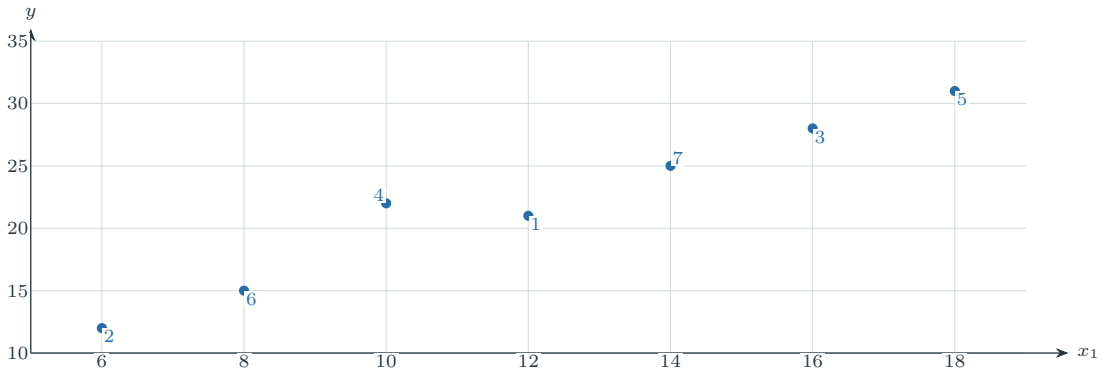
03 Prediction Plot

7 Use each house's area and room count to plot **the predictions from Question 5**
Draw circles, then connect each prediction to the same house's actual price with a **vertical line**

$P_i = (x_1^{(i)}, \hat{y}^{(i)})$ $\hat{y}^{(i)} = x_1^{(i)} + 3x_2^{(i)} + 2$

$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$	$i=7$

* Same axes and actual data as Question 3 Blue points: actual prices · Circles: predictions



Use the plot to check the error signs for Houses 1 and 4

$e^{(i)} = \hat{y}^{(i)} - y^{(i)}$ vertical distance $= |e^{(i)}|$

$e^{(1)} =$ _____ $e^{(4)} =$ _____

04 Parameter Update

$\mathbf{w} = [1, 3], \quad b = 2 \qquad \alpha = 0.01 \qquad m = 7$

8 Copy the errors from Question 5 and multiply each by the corresponding feature

$e^{(i)} = f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \qquad e^{(i)}x_1^{(i)}, \quad e^{(i)}x_2^{(i)}$

House	$e^{(i)}$	$e^{(i)}x_1^{(i)}$	$e^{(i)}x_2^{(i)}$
1			
2			
3			
4			
5			
6			
7			
Sum			

9 Calculate the gradient for each parameter

$$g_1 = \frac{\partial J}{\partial w_1} = \frac{1}{m}\sum_{i=1}^{m} e^{(i)}x_1^{(i)} \qquad g_2 = \frac{\partial J}{\partial w_2} = \frac{1}{m}\sum_{i=1}^{m} e^{(i)}x_2^{(i)} \qquad g_b = \frac{\partial J}{\partial b} = \frac{1}{m}\sum_{i=1}^{m} e^{(i)}$$

$g_1 =$ _____ $g_2 =$ _____ $g_b =$ _____

* Use the provided derivatives The gradient for b has no feature multiplier

10 Update all three values simultaneously from the same starting point

$w_1^{\text{new}} = w_1 - \alpha g_1 \qquad w_2^{\text{new}} = w_2 - \alpha g_2 \qquad b^{\text{new}} = b - \alpha g_b$

$w_1^{\text{new}} = 1 - 0.01 \times$ _____ $=$ _____

$w_2^{\text{new}} = 3 - 0.01 \times$ _____ $=$ _____ $\qquad b^{\text{new}} = 2 - 0.01 \times$ _____ $=$ _____

11 Can you update w_1 first, then use its new value to recalculate g_2?

☐ Yes ☐ No

05 Cost and Feature Scale

12 Calculate House 4's error after the update and compare it with the overall cost change

$f_{\text{new}}(\mathbf{x}) = 0.88x_1 + 2.97x_2 + 1.99 \qquad \mathbf{x}^{(4)} = [10, 3], \quad y^{(4)} = 22$

$\hat{y}^{(4)}_{\text{new}} =$ _____ $e^{(4)}_{\text{new}} =$ _____

* Overall cost across all seven houses: $J_{\text{old}} = 15/14 \approx 1.071$, $J_{\text{new}} = 11.8016/14 \approx 0.843$

☐ Lower cost means every error decreases ☐ Some errors can increase

13 For House 1, compare prediction changes when each weight is increased by 0.1 separately

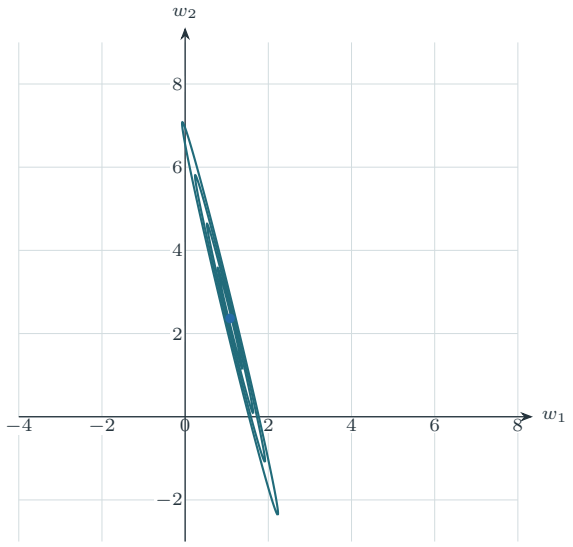
$\Delta\hat{y} = x_j \Delta w_j \qquad \mathbf{x}^{(1)} = [12, 3]$ (other values fixed)

$\Delta w_1 = 0.1 \Rightarrow \Delta\hat{y} =$ _____ $\Delta w_2 = 0.1 \Rightarrow \Delta\hat{y} =$ _____

14 The contours of our cost function are shown below

Analyze any features that could make it difficult for gradient descent to reach the minimum

* Same seven examples, $J(w_1, w_2, b = 2)$ A slice at fixed $b = 2$ · Blue point: minimum of this slice



06 Feature Scaling

15 Calculate the reference values for each feature

$$\mu_j = \frac{1}{m}\sum_{i=1}^{m} x_j^{(i)} \qquad R_j = x_{j,\text{max}} - x_{j,\text{min}}$$

Feature	Min.	Max.	Mean μ_j	Range R_j
x_1				
x_2				

16 Divide each feature by its own maximum

$$x_{1,\text{scaled}} = \frac{x_1}{18} \qquad x_{2,\text{scaled}} = \frac{x_2}{4}$$

House	$x_1/18$	$x_2/4$
1		
2		
3		
4		
5		
6		
7		

17 Apply mean normalization

$$x_{j,\text{mean}} = \frac{x_j - \mu_j}{x_{j,\text{max}} - x_{j,\text{min}}} \qquad x_{1,\text{mean}} = \frac{x_1 - 12}{12} \qquad x_{2,\text{mean}} = \frac{x_2 - 3}{2}$$

House	$(x_1 - 12)/12$	$(x_2 - 3)/2$
1		
2		
3		
4		
5		
6		
7		

18 What does a negative transformed room count mean?

☐ The house has a negative number of rooms ☐ Fewer rooms than the mean

07 Z-score and New Inputs

19 Apply z-score normalization

$$x_{j,z} = \frac{x_j - \mu_j}{\sigma_j} \qquad \mu_1 = 12, \quad \sigma_1 = 4 \qquad \mu_2 = 3, \quad \sigma_2 = \frac{2}{\sqrt{7}} \approx 0.756$$

* Standard deviation σ measures spread around the mean; use the given values
Use fractions or three decimal places For room count, you may use $1/\sigma_2 \approx 1.323$

House	$(x_1 - 12)/4$	$(x_2 - 3)/(2/\sqrt{7})$
1		
2		
3		
4		
5		
6		
7		

20 Check the equations that preserve predictions when input units change

$$u_1 = \frac{x_1}{18}, \quad u_2 = \frac{x_2}{4} \qquad x_1 + 3x_2 + 2 = 18u_1 + 12u_2 + 2$$

* The same model in different units; keeping the old weights after scaling can change predictions

Use a new house of 150 m^2 with 3 rooms $\mathbf{x} = [15, 3]$

$u_1 =$ _____ $u_2 =$ _____

$15 + 3 \times 3 + 2 =$ _____ $18u_1 + 12u_2 + 2 =$ _____

21 Transform a new house of 210 m^2 with 5 rooms using the same reference values

$$u_1 = \frac{21}{18} \qquad u_2 = \frac{5}{4}$$ use training reference values $18, 4$

$u_1 =$ _____ $u_2 =$ _____

22 Choose the correct statement

☐ Z-scores always lie between -1 and 1 ☐ Each feature uses its own mean and standard deviation

08 Scaling and Contours

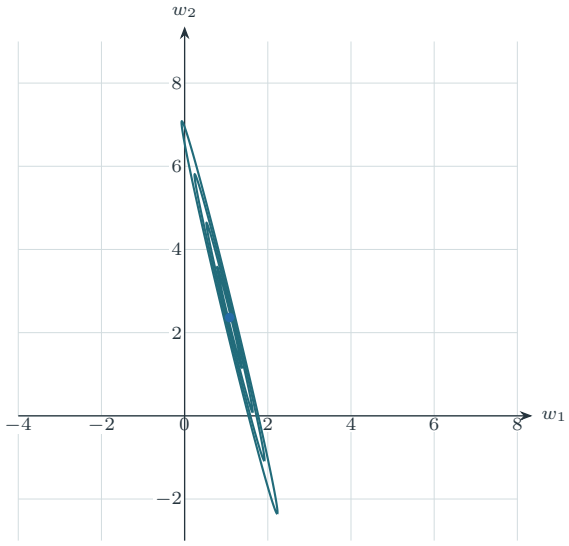
23 Compare the contours from Question 14 with those after scaling Explain any improvement

$$u_1 = \frac{x_1}{18}, \quad u_2 = \frac{x_2}{4} \qquad v_1 = 18w_1, \quad v_2 = 4w_2 \qquad b = 2$$

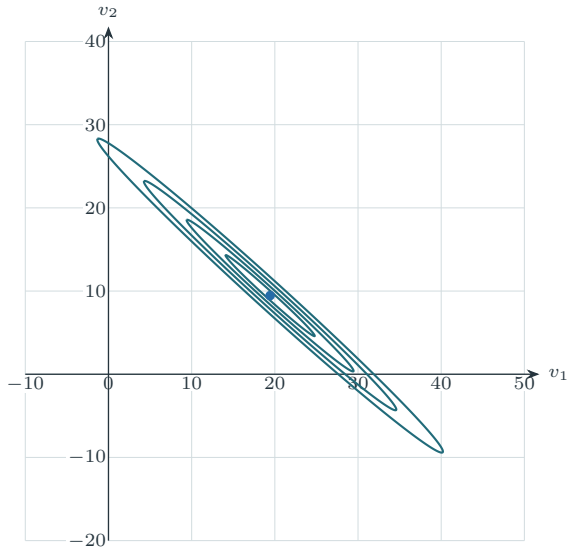
$$J_{\text{scaled}}(v_1, v_2) = \frac{1}{2 \times 7} \sum_{i=1}^{7} \left(v_1 u_1^{(i)} + v_2 u_2^{(i)} + 2 - y^{(i)} \right)^2$$

* Same houses and targets as Question 14; fixed $b = 2$ Both axes use equal scale within each plot

Q14 / Before



Q16 / After



* Both plots show costs $0.1,\ 0.35,\ 0.8,\ 1.5$ above the minimum